Power Budget

Team Number: 208

Project Name: Self Balancing Work Bench

Team Member Name: Emma Cumiford

All Major Components	Component Name	Part Name	Supply Voltage Range	Qty	Absolute Maximum Current	Total Current	Unit
	H-Bridge Motor Driver	FAN8100N	1.8–9 V	1	1500	40	mA
	Linear Solenoid Actuator	DS-0420S	3–9V	1	577.5	577.5	mA
	Curiosity Nano	PIC18F57Q43	1.8-5.1V	1	500	500	mA

5V Power Rail	Component Name	Part Name	Supply Voltage Range	Qty	Absolute Maximum Current	Total Current	Unit
	H-Bridge Motor Driver	FAN8100N	2.2–9 V	1	1500	40	mA
	Linear Solenoid Actuator	DS-0420S	3–9V	1	577.5	577.5	mA
	Curiosity Nano	PIC18F57Q43	1.8-5.1V	1	500	500	mA
					Subtotal	**1117.5**	**mA**
					Safety Margin	**25%**	
					tal Current Required on 5V Rail	**1396.875**	**mA**
Regulator	5V Regulator	L7805ABV-DG	0-35V	1	1500	1500	mA
				Total Remaining Current on 5V Rail		103.125	mA

External Power Supply	Component Name	Part Name	Supply Voltage Range	Output Voltage	Absolute Maximum Current	Total Current	Unit
Power Source Selection	Plug-in Wall Supply	179-SWI18-9-N-P5-ND	9V	9V	2200	2200	mA
Power rails connected to External Power Source	5V Regulator	L7805ABV-DG	0-35V	5V	1500	1500	mA
					Total Remaining Current Available on External Power Source	**700**	**mA**